Parent’s signature:

Name	Index Number	CTG

YISHUN JUNIOR COLLEGE
JC 2 Preliminary Examination 2011

BIOLOGY **8875/02**

Higher 1 Paper 2 **16th August 2011**

2 hours **Tuesday 1400-1600**

Additional materials: 3 pieces of Answer Paper, threads

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CTG on all the work that you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do **not** use staples, paper clips, highlighters, or glue or correction fluid.

Section A
Answer **all** questions in the spaces provided in question paper.

Section B
Answer only **one** question on the separate Answer papers provided.
At the end of the examination, fasten all Answer Papers used for answering **Section B** (essay question) securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner’s Use	
Section A	
Q1	/9
Q2	/12
Q3	/13
Q4	/6
Sub-total	**/40**
Section B	**/20**
Total	**/60**

This question paper consists of **13** printed pages.

Section A

Answer **all** the questions.

Write your answers in the spaces provided.

1. Mitochondria are important organelles in the cells. They fuel the cells through processes like oxidative phosphorylation. Mitochondria have their own DNA, distinct from the cell's own genetic material, and they multiply on their own. But in the process, mitochondria can accumulate small genetic mutations, which under normal circumstances are corrected by specialised repair systems within the cell. Over time, as we age, the number of mutations begins to outstrip the system's ability to make repairs, and mitochondria start malfunctioning and dying.

Many scientists consider the loss of healthy mitochondria to be an important underlying cause of aging in mammals. As resident mitochondria falter, the cells they fuel degenerate or die. Muscles shrink, brain volume drops, hair falls out or loses its pigmentation, and soon enough we are, in appearance and beneath the surface, old.

(a) According to the passage, the resident mitochondria "fuel" the cells. What does this "fuel" refers to? [1]

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(b) Explain the exact mechanism in which this "fuel" is produced in oxidative phosphorylation. [4]

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(c) Suggest how the lack of fuel accelerates the ageing process. [1]

(d) Suggest why mitochondria DNA are more susceptible to damages compared to DNA found within the nucleus. [1]

(e) Mitochondrial DNA has been an important tool for modern evolutionary studies. [2]

Explain why mitochondrial genes like cytochrome b are used to measure changes in DNA sequences in closely related species.

[Total: 9 marks]

2. Fig. 2.1 below shows part of an α-helix of a polypeptide chain commonly found in many types of protein.

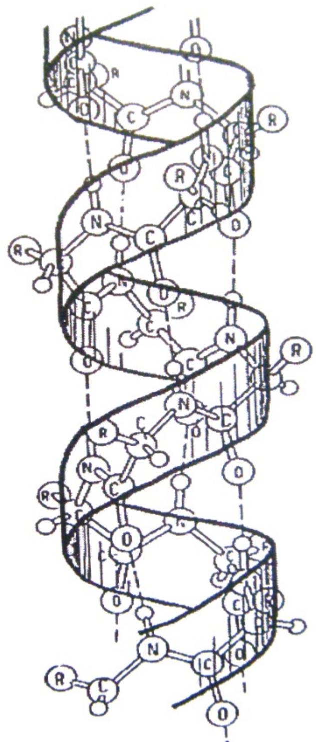

Fig. 2.1

(a) State the repeating monomer of a polypeptide chain. [1]

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(b) Explain how the α-helical structure of a polypeptide chain is maintained. [2]

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(c) Explain what would happen to the structure of the α-helix if the polypeptide chain was heated to a temperature of above 70°C. [2]

(d) α-keratins belongs to a family of fibrous structural proteins containing large amounts of the amino acid cysteine. Explain how the molecular structure of α-keratins contributes to its strength and insolubility. [2]

(e) Describe two structural features of haemoglobin that are related to its functions [2]

(f) Fig. 2.2 shows the relationship between pH and the relative activity of two different protein-digesting enzymes: trypsin and pepsin

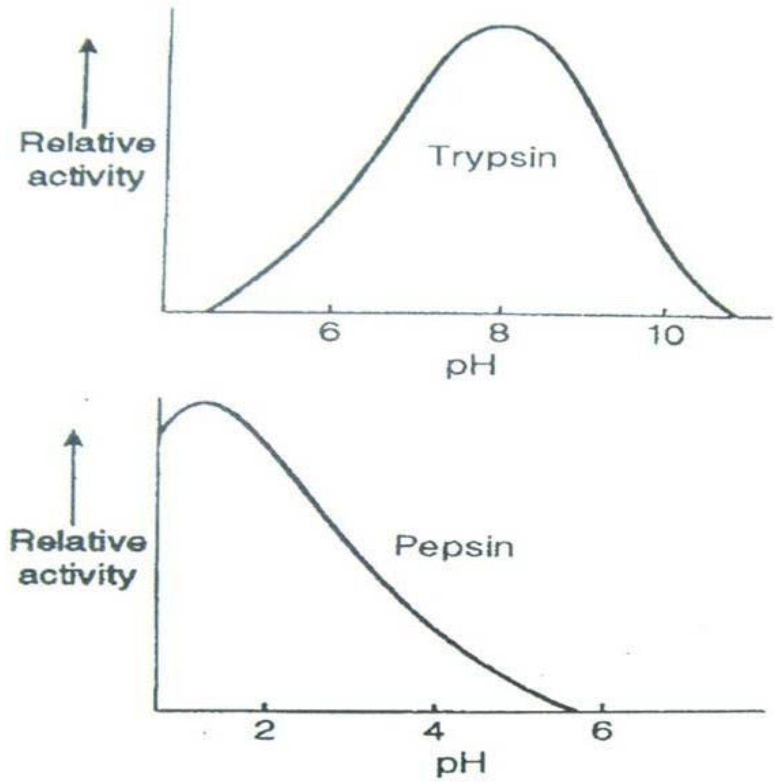

Fig. 2.2

(i) With reference to Fig. 2.2, describe and explain the effects of changes in pH on the activity of trypsin and pepsin. [2]

(ii) Suggest why different enzymes have different optimum working pH. [1]

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[Total: 12 marks]

3 The DNA of a family affected by a rare autosomal disease was analyzed using the technique of gel electrophoresis. The gene suspected to be responsible for the disease was analyzed. Fig. 3.1 shows the alleles and their respective restriction sites for *EcoR*1.

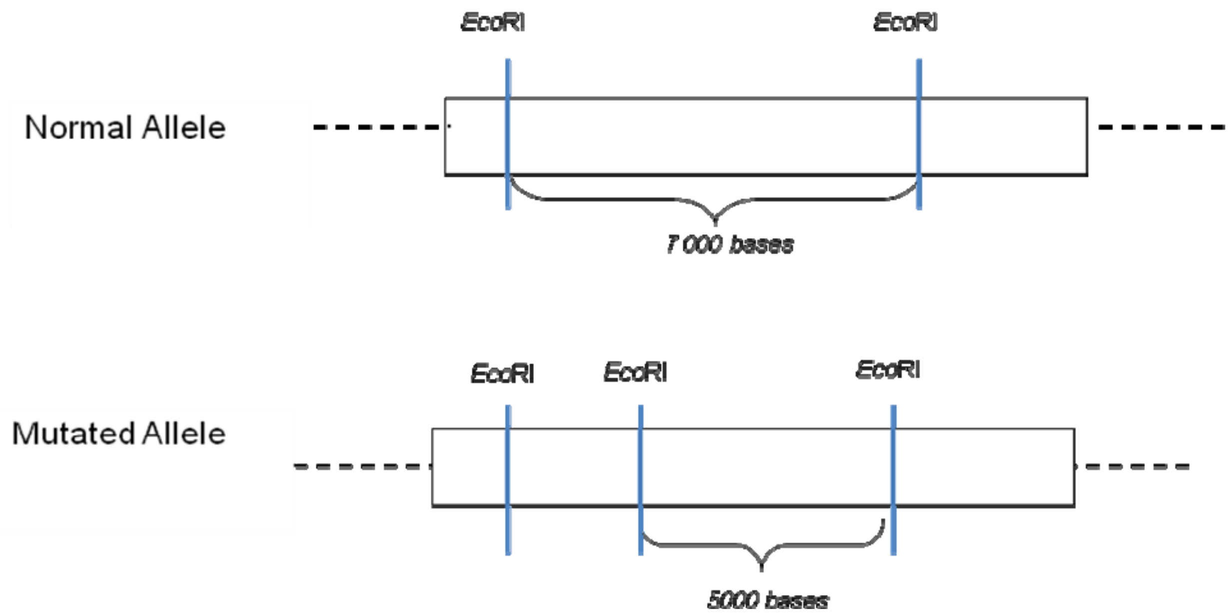

Fig. 3.1

(a) Define what is meant by an autosomal recessive disease. [2]

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(b) State the type of enzyme that will cut DNA fragments. [1]

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(c) With reference to Fig. 3.1, explain what is gel electrophoresis and how the two alleles can be distinguished using this technique. (Include in your answer how would the fragments of the 2 alleles differ) [4]

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(d) Mark on the diagram, the position of the allele in which the gene probe must bind to. [1]

Fig. 3.2 shows the pedigree chart analysis of a family affected by the disease.

A phenotypically normal couple had three children W, X and Y, with only one of them affected by the disease (indicated by the shaded square).

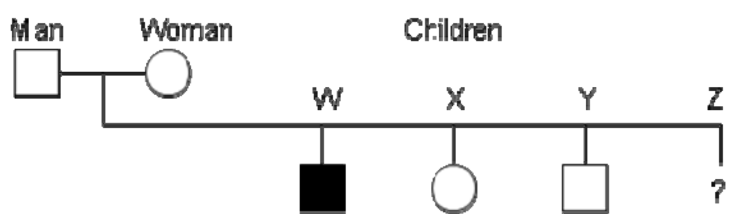

Fig. 3.2

(e) Draw a genetic diagram to explain the pedigree chart [4]

(f) State the probability that child Z is an unaffected girl. [1]

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[Total: 13 marks]

4 The haemoglobin molecules of people who are homozygous for sickle-cell anaemia clump up into long chains, distorting and weakening their red blood cells. This distortion causes severe anaemia and potentially death.

Far from being eliminated via the process of natural selection, the sickle-cell allele is carried by half the people in some areas of Africa. This distribution seems to result from the counterbalancing effects of anaemia and malaria, a disease that formally cause high death rates in equatorial Africa.

Heterozygotes enjoy some protection against malaria. Malarial parasites inside a heterozygote's red blood cells use up oxygen, causing the sickle-cell haemoglobin to clump and the cells to become sickle-shaped. Infected sickle cells are destroyed by the spleen before the parasites can complete their development. Heterozygotes, therefore, have mild anaemia, but do not succumb to malaria. Hence, in these parts of Africa, Heterozygotes survived better than either type of homozygotes and reproduced the most.

Fig 4.1 below shows the distribution of a haemoglobin's oxygen carrying capacity in the population at the beginning of the malaria epidermic

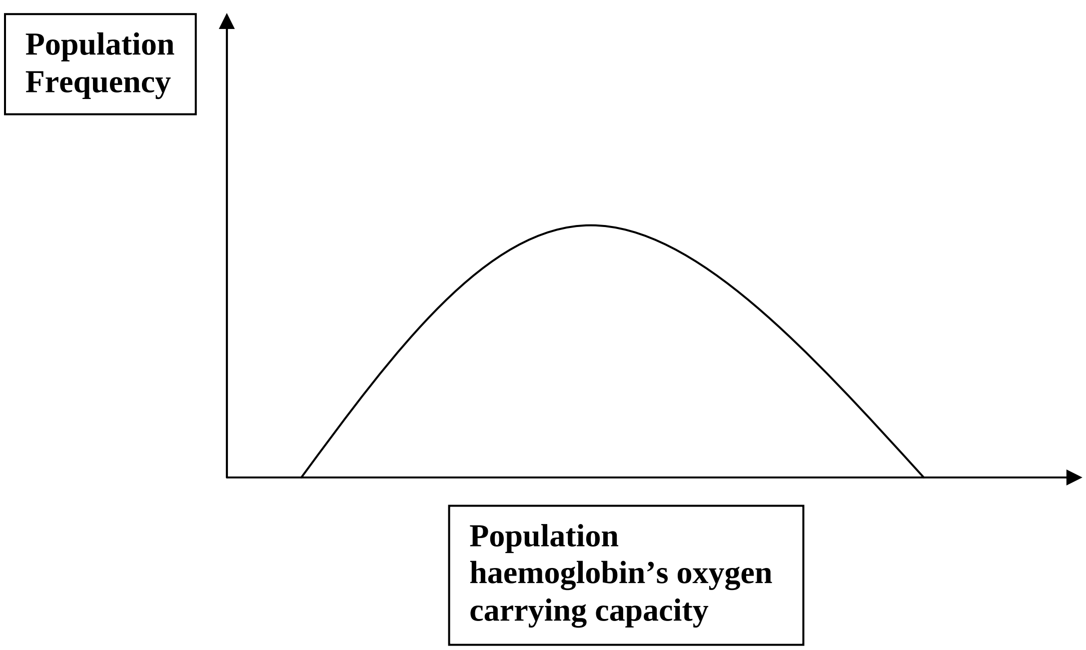

Fig. 4.1

(a) Explain the term natural selection. [2]

(b) Sketch on Fig 4.1 how the distribution of the graph would change if malaria persists in parts of Africa for the next 100 years. [1]

(c) Explain the form of selection that the population is experiencing. [2]

(d) Suggest how the distribution of the graph would change if malaria is eradicated from the affected regions of Africa. [1]

[Total: 6 marks]

Section B

Answer only **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a), (b)** etc., as indicated in the question.

5. **(a)** Discuss the advantages and disadvantages of genetic modified salmon. [6]

(b) Using appropriate examples, explain the normal functions of stem cells. [6]

(c) Explain how the light-dependent reaction during photosynthesis produces ATP and NADPH. [8]

6. **(a)** Describe the structure of the DNA molecule. [6]

(b) Explain the process of DNA replication and how it helps maintain genetic stability. [8]

(c) Compare and contrast transcription and translation. [6]

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